

A. Can you Traverse-1?

time limit per test: 5 seconds

memory limit per test: 256 megabytes

 A useful tool for constructing graphs: https://csacademy.com/app/graph_editor/

You are given an **undirected unweighted** graph with N cities and M roads. The cities are numbered from 1 to N . You may assume, the graph is connected, meaning there is a path between any pair of cities. There are no self-loops (no road connects a city to itself) and no multiple edges between the same pair of cities.

Your task is to perform a Breadth-First Search (BFS) starting from node 1 and print the order in which the nodes are visited.

Pseudocode of BFS

The breadth-first-search procedure below assumes that the input graph $G = (V, E)$ is represented using adjacency lists.

 $\text{BFS}(G, s):$

```
for each vertex u in G.V:
    u.colour = 0
```

```
Q = ∅ ;
s.colour = 1
ENQUEUE(Q, s)
```

```
while Q ≠ ∅;
    u = DEQUEUE(Q)
    for each v in G.Adj[u]:
        if v.colour == 0:
            v.colour = 1
            ENQUEUE(Q, v)
```

Input

The first line contains two integers N and M ($1 \leq N \leq 2 \times 10^5, 1 \leq M \leq 3 \times 10^5$) — the number of cities and the total number of roads.

The next M lines will contain two integers u_i, v_i ($1 \leq u_i, v_i \leq N$) — denoting there is an edge between city u_i and city v_i .

Output

Print the BFS traversal starting from node 1 as a space-separated list of visited nodes. If there are multiple BFS path traversal order, you may print any.

Examples

input

Copy

```
4 3
1 4
3 2
1 3
```

CSE221 Assignment 05 Summer 2026

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4 weeks

Contestant



→ Submit?

 Language: Java 21 64bit

 Choose file: Choose File No file chosen


Success!



Submit

output

Copy

1 3 4 2

input

Copy

```

6 10
3 1
1 6
6 4
4 5
5 2
6 2
4 3
5 6
3 6
1 5

```

output

Copy

1 3 5 6 4 2

input

Copy

```

4 5
1 3
3 4
4 2
3 2
1 4

```

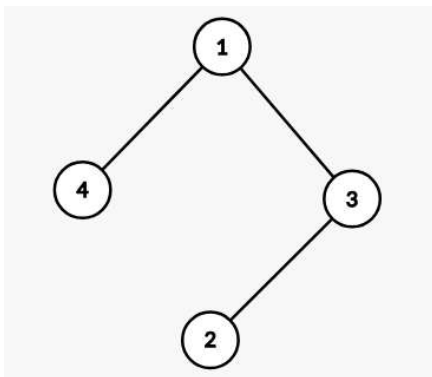
output

Copy

1 3 4 2

Note

In Sample Test Case 1, the graph looks like this:



There are two valid paths that follow the BFS order: 1 4 3 2 and 1 3 4 2. You may print either one of these paths in the output.

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